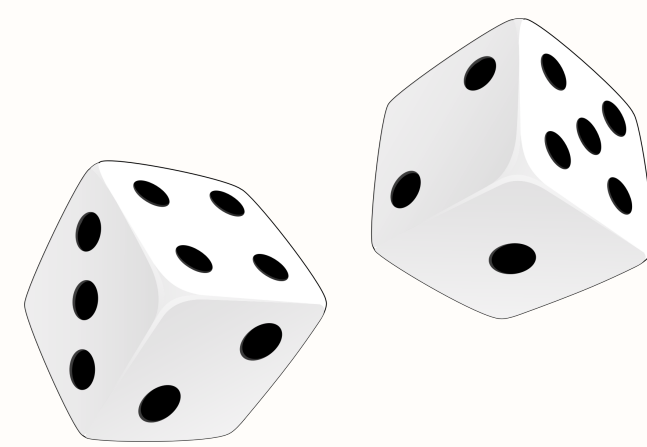




Anton Waldron¹, Dr. Alan Hegarty²,

¹BSc Financial Mathematics, University of Limerick

²Department of Mathematics and Statistics, University of Limerick

Introduction

Motivation

The motivating identity for the Multilevel Monte Carlo method is:

$$\mathbb{E}[X] = \mathbb{E}[Y] + \mathbb{E}[X - Y],$$

which can be expanded to the case where we have a random variable P and P_ℓ Monte Carlo approximations to that random variable, where as ℓ increases, accuracy improves as the computational cost worsens. This allows the use of the following equation

$$N_0^{-1} \sum_{n=1}^{N_0} P_0^{(0,n)} + \sum_{\ell=1}^L \left\{ N_\ell^{-1} \sum_{n=1}^{N_\ell} (P_\ell^{(\ell,n)} - P_{\ell-1}^{(\ell,n)}) \right\},$$

as an unbiased estimator for $\mathbb{E}[P]$ ¹.

¹ The superscript ℓ denotes that independent samples are used on each level.

The MLMC Theorem

Theorem 1. Let P denote a random variable, and let P_ℓ denote the corresponding level ℓ numerical approximation. [1] If there exist independent estimators Y_ℓ based on N_ℓ Monte Carlo samples, each with expected cost C_ℓ and variance V_ℓ , and positive constants $\alpha, \beta, \gamma, c_1, c_2, c_3$ such that $\alpha \geq \frac{1}{2} \min(\beta, \gamma)$ and

$$(i) \quad |\mathbb{E}[P_\ell - P]| \leq c_1 2^{-\alpha \ell}$$

(ii)

$$\mathbb{E}[Y_\ell] = \begin{cases} \mathbb{E}[P_0], & \ell = 0, \\ \mathbb{E}[P_\ell - P_{\ell-1}], & \ell > 0, \end{cases}$$

$$(iii) \quad V_\ell \leq c_2 2^{-\beta \ell}$$

$$(iv) \quad C_\ell \leq c_3 2^{\gamma \ell}$$

then there exists a positive constant c_4 such that for any $\varepsilon < e^{-1}$ there are values L and N_ℓ for which the multilevel estimator $Y = \sum_{\ell=0}^L Y_\ell$ has a mean-square error bounded by ε^2 with a computational complexity C satisfying

$$\mathbb{E}[C] \leq \begin{cases} c_4 \varepsilon^{-2}, & \beta > \gamma, \\ c_4 \varepsilon^{-2} (\log \varepsilon)^2, & \beta = \gamma, \\ c_4 \varepsilon^{-2 - (\gamma - \beta)/\alpha}, & \beta < \gamma. \end{cases}$$

It's important to note that in this theorem, the constants c_1, c_2 , and c_3 are assumed to be known, but in practice they are heuristically estimated using the MLMC algorithm.

References

- [1] M. B. Giles, "Multilevel Monte Carlo Methods," *Acta Numerica*, vol. 24, pp. 259–328, 2015.
- [2] M. B. Giles, "Improved Multilevel Monte Carlo Convergence Using the Milstein Scheme," in *Monte Carlo and Quasi-Monte Carlo Methods 2006*, pp. 343–358.
- [3] F. A. Longstaff and E. S. Schwartz, "Valuing American Options by Simulation: A Simple Least-Squares Approach," *The Review of Financial Studies*, vol. 14, no. 1, pp. 113–147, 2001.

Acknowledgements

I extend my thanks to Dr. Alan Hegarty for our frequent meetings and continuous feedback throughout the project. I also thank Prof. Mike Giles for the extensive publicly available MLMC code-base.

Multilevel Monte Carlo for π

We use the identity:

$$\hat{\pi} = 4 \left(\frac{\text{No. Red Points}}{\text{Total No. Points}} \right).$$

The number of points used is $f(\ell) = 8^\ell$. On each level, the fine estimate $\hat{\pi}_f$ is calculated using all available points whereas the coarse estimate $\hat{\pi}_c$ uses a subset of size $8^{\ell-1}$, the "Level 2 points" with the black outline.

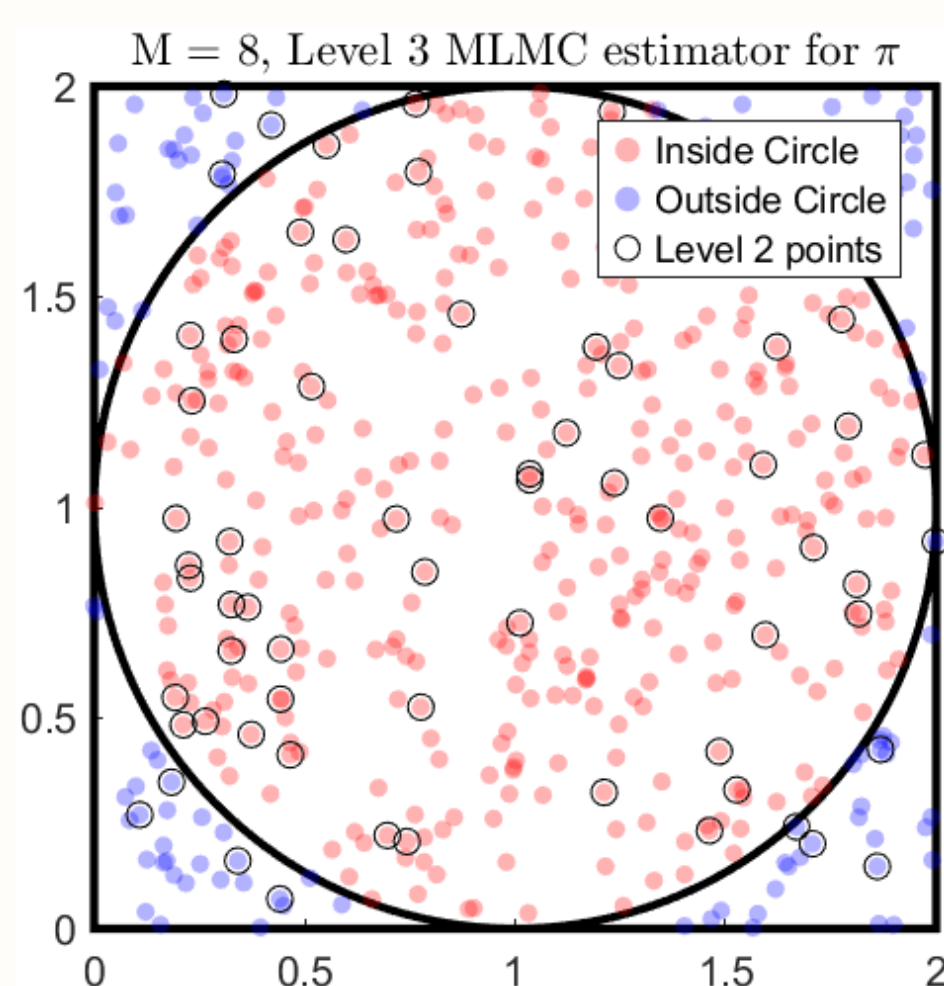


Figure 1: Illustration of MLMC for π

We find that $|\mathbb{E}[P - P_\ell]| \approx O(2^{-0.9\ell})$, $V_\ell \approx O(2^{-3\ell})$, and $C_\ell \approx O(2^{3\ell})$. There are only cost savings for RMSE targets of 5×10^{-3} and 1×10^{-2} , for higher RMSE targets the cost is higher for MLMC.

MLMC for Options Pricing

We assume that the stock price process is log-normally distributed with a constant interest rate r and volatility σ . Using the Milstein discretisation for the Black-Scholes SDE we simulate the stock price process as:

$$\hat{S}_{n+1} = \hat{S}_n (1 + r\Delta t + \sigma \Delta W_n + \frac{1}{2} \sigma^2 (\Delta W_n)^2)$$

where the ΔW_n term refers to the Brownian increment which has variance Δt and mean 0. To ensure the paths are coupled (correlated); when computing the coarse path we use a sum of the Brownian increments used to compute the fine path. An illustration of this concept is seen below:

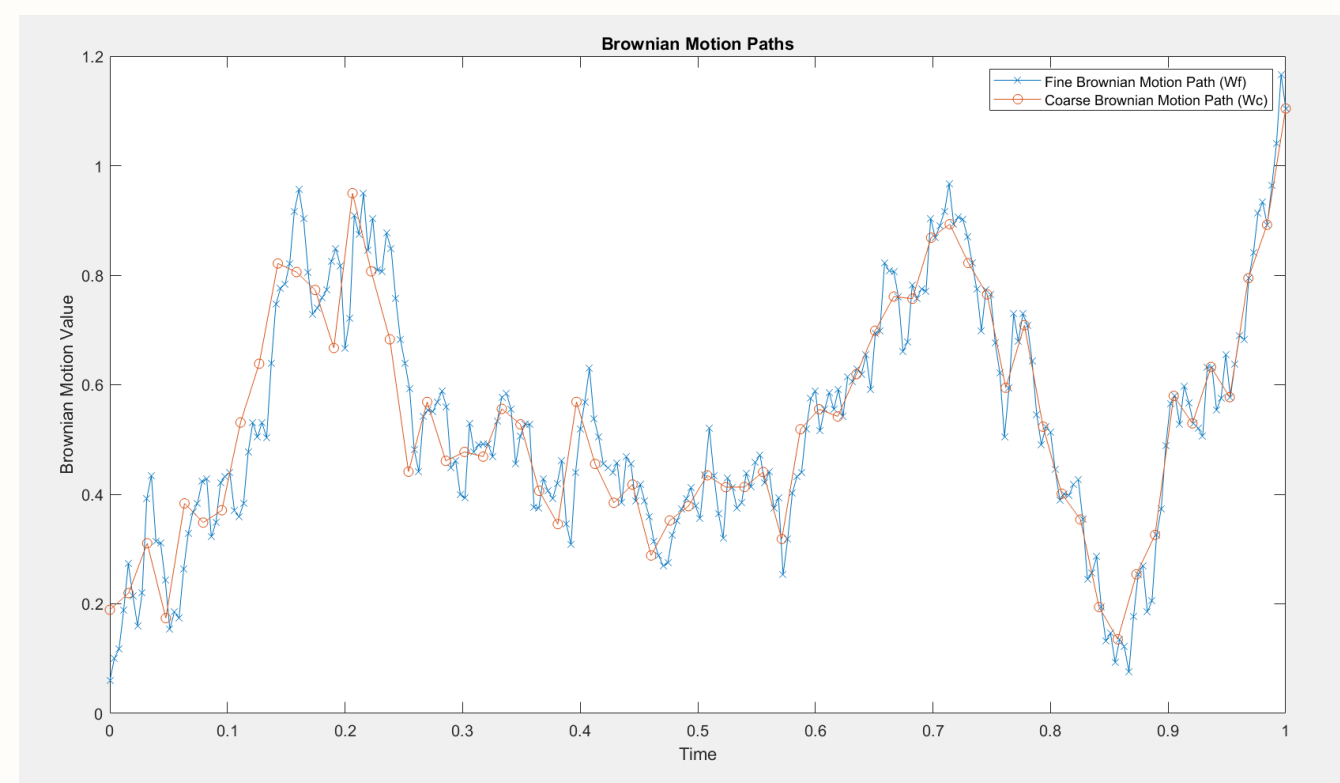


Figure 2: Illustration of coupling Brownian Motions; fine path in orange; coarse path in blue.

For a European standard call option we aim to determine

$$\tilde{\mathbb{E}}[\max(S_T - K, 0)],$$

and the results using MLMC and the Milstein discretisation are that $|\mathbb{E}[P - P_\ell]| \approx O(2^{-0.96\ell})$, $V_\ell \approx O(2^{-1.92\ell})$ and $C_\ell \approx O(2^\ell)$.

The cost savings are that MLMC is approximately 221 times cheaper for an RMSE target of 5×10^{-3} , 111 times cheaper for an RMSE target of 1×10^{-2} and has double digit savings for any higher RMSE targets.

Fixed Lookback Options

We aim to determine

$$\tilde{\mathbb{E}}[\max(S_{\max} - K, 0)].$$

To estimate the maximum we assume the stock price is a Brownian Bridge between timesteps and use known properties to calculate the maximum. Coupling is achieved by using summed Brownian increments and an interpolant $\hat{S}_{n+1/2}$ for the coarse terms.

The result is $|\mathbb{E}[P - P_\ell]| \approx O(2^{-0.82\ell})$, $V_\ell \approx O(2^{-1.8\ell})$, and $C_\ell \approx O(2^\ell)$.

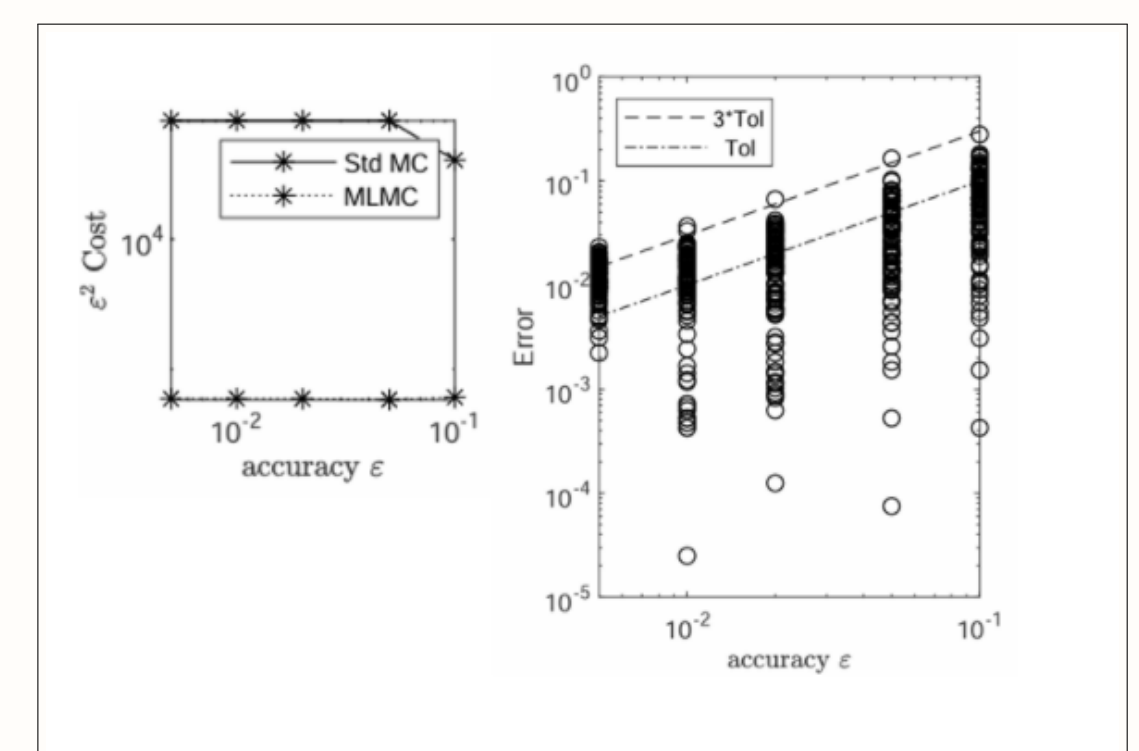


Figure 3: Results and Cost for Fixed Lookbacks

MLMC results in a cost that is approximately 136 times cheaper for RMSE targets from 5×10^{-3} to 5×10^{-2} and 66 times cheaper for 1×10^{-1} .

American Puts

We aim to determine

$$\tilde{\mathbb{E}}[(1 + r)^{-\tau} \max(K - S_\tau, 0)],$$

where τ is an optimal stopping time. Fixed Lookback puts can be thought of as a solution to the dual problem of American options, calculating an upper bound for their price. If we adapt the Least Squares Monte Carlo approach to MLMC, we have a MLMC approach for a lower bound on the price [3]. The result is that the MLMC parameters are estimated as $|\mathbb{E}[P - P_\ell]| \approx O(2^{-1.45\ell})$, $V_\ell \approx O(2^{-0.34\ell})$, and $C_\ell \approx O(2^{2\ell})$. The cost savings are that MLMC is approximately 2 times cheaper for RMSE targets of 2×10^{-2} and 1×10^{-2} , and 1.5 times cheaper for 5×10^{-2} . Unfortunately we were unable to assess if this trend continues for lower RMSE targets due to computational constraints.

Summary

We begin by introducing the Multilevel Monte Carlo (MLMC) method with a simple application for estimating π . Following this, we move to a more suitable setting for MLMC - options pricing in computational finance. In this setting we find that for fixed lookback options, which provide an upper bound for American options, we achieve savings up to $166\times$ cheaper than standard Monte Carlo. In the case of a lower bound, we adapt the LSM approach to MLMC and find that it is $2\times$ cheaper for RMSE targets of 2×10^{-2} and 1×10^{-2} .